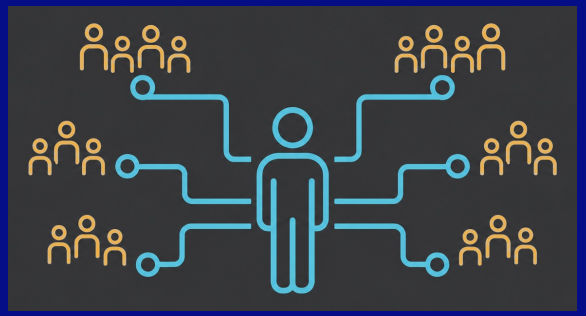


DHUM101
Socially Responsible Computing

University of Otago
ŌTĀKOU WHAKAIHU WAKA
Te Kura Rorohiko
School of Computing
Te Reo Inarahi me te Mātai Wetereo
English and Linguistics

Lecture 1
Introduction to DHUM101

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  Semester 2, 2026
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What is this paper about?

Why "DHUM"

- DHUM is the paper code for Digital Humanities
- "Humanities are academic disciplines that study aspects of human society and culture, including certain fundamental questions asked by humans." (Wikipedia)
- We hope to convince you that:
  - Computing technologies have widespread impacts on individuals and societies
  - We can't leave it to the technologists to predict and prevent or mitigate negative impacts.
  - We need a dialogue involving the creators of new technologies, humanities scholars and the general public.

- Teaching team across Sciences and Humanities:
  - Stephen Cranefield, School of Computing
  - Hunter Hatfield, English and Linguistics





To get you thinking (1)

Surveillance pricing "uses algorithms, powered by artificial intelligence or other data-processing techniques, to set custom prices or wages based on the collection of an individual's information. This data could include everything from where an individual lives and what they have bought in the past, to their financial status, travel habits and affiliations".[1]

Surveillance wage-setting (proposed for the gig economy) "works the same way as surveillance pricing: through analysis of personal data about a person's financial history, personal associations, demographics, and behaviors".[2]

☹️ "Critics of surveillance pricing say that companies exploit this data to charge buyers the most that they are willing to pay, and give workers the lowest amount they are willing to accept". Also it can be seen as a breach of privacy.

😀 Economic theory suggests that individualised pricing and wage-setting increases the efficiency of allocation of goods and jobs to people, i.e. more people may get an acceptable outcome.

Join at:
vevox.app

ID:
118-398-771

1. https://www.theguardian.com/technology/2026/jun/03/colorado-governor-veto-ban-surveillance-pricing,
2. https://cdt.org/insights/op-ed-clocked-in-how-surveillance-wage-setting-can-affect-people-with-disabilities/

To get you thinking (2)

In 2024, the Austrian Social Democratic Party (SPÖ) introduced parliamentary motions calling for a legal "right to an analog life". The proposals included:

- a right to receive paper bills without extra charges,
- ensuring that government benefits and services remain accessible without the internet,
- preventing citizens from being forced to use digital identity systems such as ID Austria.

Since then, this issue has become a recurring topic in Austrian political debate.

Meanwhile, in New Zealand, banks have been closing branches, the number of NZ Post outlets is reducing and Dunedin has (e.g.) no Department of Internal Affairs office where you can apply for a passport or get a birth certificate or marriage licence.

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What is Responsible Computing?

Responsible computing involves the ethical design, development, and use of technology. It emphasizes minimizing harm, respecting user rights, and fostering sustainability. This approach ensures that technology benefits society without creating unintended consequences.

Core Tenets of Responsible Computing

- Ethical Decision-Making: Considering the societal impact of technology at every stage of development.
- Inclusivity: Designing systems accessible to all, regardless of ability, location, or resources.
- Transparency: Ensuring users understand how systems work and how their data is used.
- Sustainability: Reducing the environmental impact of computing through energy-efficient systems.
- Accountability: Holding developers, organizations, and users responsible for ethical breaches.

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Why it Matters?

- For Developers
  Builds trust in technology and fosters user loyalty.
- For Society
  Protects vulnerable populations from harm.
- For the Environment
  Promotes eco-friendly practices in the tech industry.

The Role of Students and Professionals

- Stay informed about ethical practices.
- Advocate for responsible policies in your organizations.
- Develop skills to design systems that prioritize inclusivity and transparency.

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Potential social impacts of IT professionals

- “Computing professionals' actions change the world. To act responsibly, they should reflect upon the wider impacts of their work, consistently supporting the public good.”

ACM Code of Ethics and Professional Conduct, https://www.acm.org/code-of-ethics

- “[A] small fraction of humanity is currently engaged in the design and development of a set of technologies that are already transforming the everyday lives of just about everyone else” [i.e. artificial intelligence]

Floridi et al, AI4People—An Ethical Framework for a Good AI Society: Opportunities, Risks, Principles, and Recommendations, https://doi.org/10.1007/s11023-018-9482-5

- “The decisions made by engineers often have ethical dimensions and implications. Engineers develop and implement technologies that influence and shape the way we live, at times in unanticipated ways.”

Smith et al. The Responsibilities of Engineers, https://doi.org/10.1007/s11948-013-9463-2

Some prominent voices

- "In many cases within the digital context, control over platforms, infrastructure, data and computing power does not rest with States, but with major economic and technological actors. These entities effectively set the conditions for access, determine the rules of visibility and shape the very possibilities for participation. When such power is concentrated in the hands of a few, it tends to become opaque and evade public oversight, increasing the risk of distorted forms of development that give rise to new dependencies, exclusions, manipulations and inequalities."

Pope Leo XIV, https://www.vatican.va/content/leo-xiv/en/encyclicals/documents/20260515-magnifica-humanitas.html

- "The machinery that makes [AI] possible is the work of math and programming and science. But what character we choose, how it interacts with the world, how it ought to interact with the world—these are more clearly questions for the humanities, for religion, for philosophy, for society at large."

Anthropic co-founder Chris Olah, https://www.anthropic.com/news/chris-olah-pope-leo-encyclical

- "If you're making some widget, and it's probably not going to change the world, then maybe you don't need a moral philosopher. But if you take AGI [artificial general intelligence] seriously, then I can't really see how you wouldn't consider this sort of thing as important."

DeepMind (now part of Microsoft) co-founder Shane Legg (a kiwi!) om hiring a philosopher
https://www.theguardian.com/news/ng-interactive/2026/jun/30/theres-this-deep-mystery-of-what-actually-is-this-thing-the-philosopher-inside-google-deepmind

Examples of taking responsibility

A software developer decides not to collect extra user data

A developer working on a mobile app proposes collecting precise location data because "it might be useful later for analytics.

After reviewing privacy regulations such as the General Data Protection Regulation (GDPR) and the principles of data minimization, the developer changes their recommendation. Instead of collecting location continuously, the app collects it only when needed for a specific feature.

The ethical reasoning is that collecting data merely because it might become useful later increases privacy risks without providing a current benefit to users.

A data analyst removes identifying information from a dataset

A university researcher wants a data analyst to provide student records for a project. Initially the analyst plans to provide names and student IDs because it would be convenient.

After consulting privacy legislation and institutional ethics guidelines, the analyst decides to provide a de-identified dataset instead.

This is a common example of balancing research utility against the principle of protecting individuals from unnecessary disclosure.

Examples of taking responsibility

A system administrator refuses to access employee emails

A system administrator has technical access to all staff email accounts. A manager asks the administrator to "have a quick look" through an employee's emails because the manager suspects poor performance.

The administrator reviews organizational policy and privacy obligations and concludes that there is no authorized business process for such access. The request is declined and referred to Human Resources and legal staff.

The ethical concern is misuse of privileged access. Many professional codes emphasize that access rights should be exercised only for legitimate and authorized purposes.

A software engineer objects to a "dark pattern"

A product team wants to make the "unsubscribe" button difficult to find in order to reduce customer churn.

An engineer raises concerns after reviewing professional obligations concerning honesty and user autonomy, such as those found in the Association for Computing Machinery Code of Ethics.

The team ultimately adopts a clearer opt-out process.

Examples of taking responsibility

A database administrator declines to share customer data internally

A marketing department asks for access to a customer database originally collected for service delivery.

After checking the company's privacy obligations and consent notices, the administrator realizes that customers did not consent to this secondary use.

The request is modified so that only aggregated statistics are shared.

A machine-learning engineer questions bias in a model

An engineer notices that a hiring-screening model performs worse for applicants from certain groups.

After consulting organizational AI ethics guidelines, the engineer treats it as an ethical issue and recommends additional testing and model redesign before deployment.

The system is delayed until the concerns are investigated.

Examples of taking responsibility

Workers at Google's DeepMind vote to unionize to pressure the company (2026)

Workers at DeepMind, Google's London-based AI research laboratory, voted to unionize. They want Google to commit to not developing technology "whose primary purpose is to cause harm or injury to people", to establish an independent ethics oversight body, and to grant workers the individual right to refuse to contribute to projects on moral grounds.

In 2025, Google dropped a pledge not to develop militarized AI and have provided Israeli military with increased access to their AI tools from the early days of the war in Gaza.

One worker was quoted as saying "I have joined the union due to concerns about AI being used to empower authoritarianism, whether through military or surveillance applications, both foreign and domestic. ... By unionizing, we are taking the traditional route for workers to organize and have a say."

https://www.theguardian.com/us-news/2026/may/04/google-deepmind-uk-workers-union

Examples of taking responsibility

Current and former OpenAI publish an open letter calling for greater transparency and more protections for whistle-blowers in leading AI companies (2024). They requested an end to using nondisparagement and nondisclosure agreements at OpenAI and other A.I. companies, supporting a culture of open criticism and establishment of a reporting process for employees to anonymously raise safety-related concerns. One signatory commented that "When I signed up for OpenAI, I did not sign up for this attitude of 'Let's put things out into the world and see what happens and fix them afterward". https://yonkersobserver.com/openai-whistleblowers-describe-reckless-and-secretive-culture/

Microsoft tightens human rights measures after inquiry into Israel's use of its tech (2026)

After revelations about the Israeli military's use of Microsoft cloud technology for the mass surveillance of Palestinians, Microsoft announced that their Israel operations will be temporarily managed by Microsoft France. After an internal inquiry Microsoft announced that it would tighten human-rights controls when working with national security agencies. Microsoft's vice-president stated that "We do not provide technology to facilitate mass surveillance of civilians".

https://www.europesays.com/france/14579/
https://www.theguardian.com/technology/2026/jun/04/microsoft-to-tighten-human-rights-measures-after-inquiry-into-israels-use-of-its-tech

Example of government policy-making



"AI-led innovation is a pathway to achieving national goals, such as socio-economic development, global competitiveness, and resilience. Therefore, AI governance frameworks should actively encourage adoption and serve as a catalyst for impactful innovation. That said, innovation should be carried out responsibly and should aim to maximise overall benefit while reducing potential harm. All other things being equal, responsible innovation should be prioritised over cautionary restraint."

The choice between being pro-innovation guidelines based and regulation-based (e.g. the EU AI Act) is a key difference between different countries' AI policies. NZ has a foot in both camps (regulation for the public sector and guidelines for industry).

IndiaAI Safety Institute's Guidelines "that ground India's AI governance philosophy"

Examples of harm from irresponsible IT use



https://www.youtube.com/watch?v=wA3dJcu3a2U (3:45)

Takeaway message

Two quotes from Cory Doctorow, *The Reverse Centaur's Guide to Life After AI : How to Think About Artificial Intelligence Before It's Too Late,* 2026. https://research-ebsco-com.ezproxy.otago.ac.nz/plink/980ef470-b077-3d12-9a70-4a5b47ace761

> "[T]ech bosses ... claim total authority over our possibilities, both present and future .... insisting that their abusive behavior is out of their hands, that they are merely acting in accord with some kind of iron law or great force of history. [Former UK Prime Minister] Margaret Thatcher was the world champion of this: she insisted that her cruel program of austerity and privation was inevitable, repeating the phrase 'There is no alternative'. ...
>
> Call this philosophy inevitabilism, the insistence that there is only one conceivable way to do things, and any problems you're experiencing aren't anyone's fault, they're just inescapable reality. The fact that the world is filled with sneaky Bluetooth sniffers that track your location as you walk through shopping centers or even just down the street isn't something you can blame a person or a company for; it's just a fact."
>
> "[The] thing is this: the future is up for grabs. It is not inevitable. AI isn't a genie that can't be put back into a bottle. How we use AI is up to us. Whether we use AI is up to us. The future can be ours, if we never stop remembering that the most important fact about a technology isn't what it does, it's who it does it for, and who it does it to."

And this applies to technology in general - not just AI

Teaching and assessment details

Learning outcomes

Students who successfully complete this paper will be able to:

1. Recall and discriminate between theories, principles and frameworks underlying the ethical and socially responsible development and use of digital technologies.
2. Identify and appraise diverse viewpoints of affected parties on the ethical, social, and environmental impact of selected current and emerging digital technologies.
3. Debate and apply the effects of technology company business models and the dominance of Big Tech on users as well as options for mitigation.
4. Identify and curate credible sources concerning the promise and reality of digital technologies.
5. Install, use and reflect on a selection of digital tools (e.g. apps) for social good or mitigating effects of Big Tech business models.

Teaching information

- There will be a 50 minute lecture each week at 1pm on Monday.
- You will be streamed for a tutorial (50 min.) and a lab (1 hr. 50 min.) each week *except for the first and last weeks of the semester*. See eVision for your tutorial and lab streams.
- Labs are held in a computer lab but do not involve coding.
- In the final week of the paper there will be a test held in a computer lab at a fixed time for all students.
- There is no final exam.
- Course materials will be made available in Aoroa.

Teaching schedule

Dates	Lecture	Lecturer	Tutorial	Lab	Assessment due
13-17 July	1. Paper Introduction	Stephen and Hunter			
20-24 July	2. Business, philosophical, and political drivers of the IT industry	Stephen	1	1	
27-31 July	3. Argumentation and Critical Thinking	Hunter	2	2	
3-7 Aug.	4. Taking responsibility for technological development	Guest: Holger Regenbrecht	3	3	Structured report
10-14 Aug.	5. Individual needs of consumers	Hunter	4	4	
17-21 Aug.	6. Social justice and rights	Hunter	5	5	
24-28 Aug.	7. Environmental impacts and sustainability	Stephen	6	6	
31 Aug.-4 Sept.	*Mid-semester break*				
7–11 Sept.	8. Culture and Identity	Stephen	7	7	Essay
14–18 Sept.	9. Ethical Frameworks	Hunter	8	8	
21–25 Sept.	10. Technology for good	Stephen	9	9	
27 Sept.–2 Oct.	11. Imagining futures	Hunter	10	10	Blog on Apps for good
5-9 Oct.	12. Interview with a practitioner	Stephen	11	11	
12-16 Oct.	13. Revision				Test

Assessment

- Lab Exercise Completion: 15%
  - **Due Date:** After each lab. Date and time to be confirmed.
  - Best 7 of 11 lab marks out of 2, plus a one-off bonus mark for extra effort on at least one lab.
- Structured Report: 30%
  (*Read and comment upon an argument concerning the impacts of digital technologies from a thought leader. 1,500 words*)
  - Due Date: Week 4. Date and time to be confirmed
- Essay: 25%
  (*Discuss differing impacts of an emerging technology utilizing taught frameworks. 1,500 words*)
  - Due Date: Week 8. Date and time to be confirmed.
- Blog: 10% (*Reflection on using apps for good. 500 words*)
  - Due Date: Week 11. Date and time to be confirmed.
- **Test:** 20% (*Online test on foundational knowledge; 1 hour*)
  - Date: Final week. Monday 6-7pm. 12 October
- There is no final exam

Class Reps wanted for student voice

Are you...

Proactive, friendly and keen to contribute to your learning environment?

A great communicator who wants to represent their peers?

Wanting to support the University in offering the best student experience?

student voice class reps

ousa student support

Class Reps wanted for student voice

What's in it for you?

FREE support, volunteering opportunities and events only offered to OUSA volunteers

A reference letter and certificate from OUSA for your CV
Professional development training and opportunities?

student voice class reps

ousa student support

Class Reps wanted for student VOICE

Volunteer to represent your class today!

Talk to your lecturer about signing up today

Once you are confirmed as the rep for your paper
please sign up at: **classrep.ousa.org.nz/register.php**

If you have any concerns or issues throughout the semester please
email the Student Voice Coordinator at **leda@ousa.org.nz**

student VOICE class reps

ousa student support

Thank you for your attention